Seetha Mahalakshmi Kolati

System Validation / V&V Lead – ADAS, Infotainment, ISO 26262 – 7.5+ Years

seethakolati2497@gmail.com linkedin.com/in/seetha-kolati-81293216b +91 8008713059

SUMMARY:

Automotive System Validation & V&V Lead with 7.5+ years delivering ADAS, Infotainment, and V2X validation for global OEM platforms including **Nissan Motors Limited** and **Jaguar Land Rover**.
Specialized in **defining and driving validation strategy**, coverage sufficiency, and risk-based release decisions across multi-ECU ADAS systems. Offshore validation lead coordinating multi-team execution, owning ISO 26262 SWE2/3/4 validation artifacts, ADAS white-box coverage governance, and HIL validation for release readiness, including 3.25 years onsite coordination in Japan. Hands-on experience in ADAS HIL validation including scenario execution, signal-level debugging, and integration-level analysis.

VALIDATION LEADERSHIP HIGHLIGHTS:

- Offshore validation lead for multi-team ADAS programs across perception and integration layers.
- Governed coverage reviews and CA triage before onsite submission, improving validation readiness and reducing late-cycle rework.
- Led migration of 2500+ ADAS automation scenarios into HIL automation framework, enabling scalable regression and faster defect discovery.
- Coordinated Japan onsite stakeholders for validation planning, issue closure, and release readiness tracking.
- Quality-gate reviewer for coverage sufficiency, risk justification, and validation completion criteria.

CORE SKILLS:

- Automotive System Validation & Verification
- ADAS Software Validation & Scenario Governance
- White-box Testing & Coverage Analysis (C0, C1, MC/DC)
- Integration & HIL Validation
- Coverage Sufficiency & Justification
- Risk Assessment & CA Triage
- Validation Strategy & Quality Gates
- Offshore–Onsite Coordination
- Test Reporting, KPIs & Program Reviews
- ISO 26262 (SWE2 / SWE3 / SWE4)

TOOLS, LANGUAGES & ENVIRONMENT

Validation & Test Tools:

- Vector CANoe, CANalyzer, CANape, CoverageMaster winAMS, Trace32, DDT2000, GHS Multi Compiler

Protocols & Interfaces:

- CAN, LIN, Automotive Ethernet, Signal-based communication

Validation Environments:

- Vehicle-level and System-level HIL setups, ADAS scenario-based testing

Requirements & Tracking:

- DOORS, Jira, Git

Programming & Analysis:

- Embedded C code understanding for white-box testing, Python

Process & Safety:

- ISO 26262 SWE2/3/4 artifact development, ASPICE-aligned validation processes, Coverage analysis & risk-based validation strategy, Agile

PROFESSIONAL EXPERIENCE:

Tata Consultancy Services – India | May 2018 - Present
Automotive System Validation / ADAS Validation Engineer
Client: Nissan Motors Limited (via TCS Delivery) - ADAS Programs

- Led white-box unit and integration validation planning for ADAS software across multiple vehicle generations, targeting C0, C1, and MC/DC coverage goals.
- Performed coverage analysis to identify untested logic paths, improving coverage levels and closing critical validation gaps before milestone reviews.
- Acted as quality gate reviewer for junior engineers' results, ensuring coverage sufficiency and correctness prior to onsite submission.
- Authored coverage justifications for unreachable code paths, differentiating design constraints from test gaps for audit readiness.
- Performed integration-level debugging in HIL environments using signal, log, and timing analysis to validate ADAS behavior and resolve false CAs.
- Validated ADAS features such as FEB, NCAP scenarios, LKA, and TJP, ensuring functional accuracy, safety compliance, and real-time performance.
- Performed closed-loop HIL testing with fault injection, UDS/DTC diagnostics, robustness checks using Vector tools to assess system behavior under fault conditions.
- Defined validation scope, coverage approach, and integration test strategy aligned with functional and safety requirements.
- Japanese language proficiency – JLPT N4 – used for daily client coordination and technical discussions.

Offshore SPOC – ADAS Automation & Validation Support

- Led offshore validation coordination as SPOC, driving migration of 2500+ ADAS automation scenarios into HIL frameworks and ensuring execution alignment across teams.
- Coordinated offshore–onsite validation execution, tracking progress, risks, and technical blockers across multiple teams.
- Acted as a first-level decision point for validation of framework issues, assessing downstream impact before escalation.
- Maintained structured issue tracking and collaborated with development teams to resolve framework and deployment challenges.
- Prepared weekly KPIs, dashboards, and validation status reports for customer and management reviews.
- Received On-The-Spot Award for critical support during the white-box validation milestone.

System Validation Engineer – Infotainment & V2X

- Performed system-level validation of infotainment features against functional requirements in integrated vehicle environments.
- Conducted V2X validation focused on interaction scenarios between vehicle systems and external entities, emphasizing correctness and timing.
- Supported defect analysis and cross-functional coordination with development teams.

FUNCTIONAL SAFETY

- Contributed to ISO 26262-aligned validation activities through preparation and review of SWE2, SWE3, and SWE4 artifacts.
- Ensured traceability between requirements, test cases, coverage evidence, and safety lifecycle expectations.
- Supported audits and reviews with coverage justification and validation evidence.

CERTIFICATIONS

- ISTQB Certified Tester Foundation Level (2025) issued by International Software Testing Qualifications Board
- JLPT N4 – Japanese Language Proficiency Test conducted by Japan Foundation & JEES